

Predicting Domestic Box Office Revenue

Using Time Series Analysis

The Role of Opening Weekend Performance

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Abstract

This paper studies how well early box office performance predicts total domestic revenue for movies released in 2023. Using a sample of 30 films and over 2,300 daily observations, I estimate a log-log cross-sectional model linking opening weekend revenue to total gross, and then turn to the daily revenue series to model its trend, seasonality, and short-term persistence. The cross-sectional model explains about 91% of the variation in total revenue, and the elasticity of total revenue with respect to opening weekend is close to one. Adding franchise, budget, and runtime controls does not change this result as once opening weekend is included, these variables are not significant. I also compare the predictive power of Day 1, Week 1, Week 2, and Week 3 revenue individually, and find that Week 2 is actually the single strongest predictor in-sample, though all four are extremely close. For the time series part, I remove trend and day-of-week seasonality through regression (instead of differencing, which inflates variance), and then model the residuals with AR and ARMA specifications. AR(1) is selected by both AIC and BIC. A combined regression-plus-AR(1) model produces reasonable out-of-sample forecasts on two held-out films.

1. Introduction and Motivation

Opening weekend performance is treated as a decisive signal of a film's commercial trajectory by studios, the press, and audiences. This project examines that claim econometrically. The topic is well-suited to Econ 423 because it draws on both cross-sectional regression and time series modeling, and because publicly available box office data make replication straightforward.

The main research question is:

To what extent can early box office performance be used to predict total domestic revenue?

I also look at a secondary question about franchise films and a forecasting objective for daily revenue:

Secondary: Do franchise films convert opening weekend revenue into total revenue more efficiently than non-franchise films?

Forecasting objective: To what extent can daily box office revenue be modeled and forecasted using trend, seasonality, and autoregressive dynamics?

2. Related Literature

Three papers motivate this project.

De Vany and Walls (1999) show that movie revenues follow a heavy-tailed (Pareto-like) distribution and that small differences in early conditions can produce very large differences in final outcomes. This motivates two expectations: first, that the raw data will exhibit pronounced skewness, and second, that early performance should contain substantial predictive information about final outcomes.

Eliashberg, Elberse and Leenders (2006) survey forecasting models in the film industry and point out that opening performance is the most commonly used and most powerful early signal in most practical models. This directly motivates the use of opening weekend revenue as the main predictor in the cross-sectional regression.

Walls (2005) extends the heavy-tail argument and shows that even though outcomes are very hard to predict in general, including early performance data substantially reduces forecast uncertainty. This frames the problem appropriately: box office outcomes are noisy, but early signals materially reduce forecast uncertainty.

Together, these papers justify (a) using log transformations to handle heavy tails, (b) using opening weekend revenue as the primary predictor, and (c) expecting strong but imperfect predictive power.

3. Data

I collected data from The Numbers (www.the-numbers.com) for 30 wide-release films from 2023. Two datasets were built:

- Movie-level dataset (30 observations): movie title, release date, opening weekend revenue, total domestic revenue, franchise indicator (1 if part of a series, 0 otherwise), production budget (in millions of USD), and runtime (in minutes).
- Daily-level dataset (2,300 observations): daily domestic gross for each movie from release day onward, with a "days since release" counter.

Films remain in theatres for varying lengths of time, but the majority of box office activity happens in the first 6–8 weeks, so the daily analysis is restricted to the first 50 days of each film's run. This ensures comparability across films.

4. Descriptive Statistics

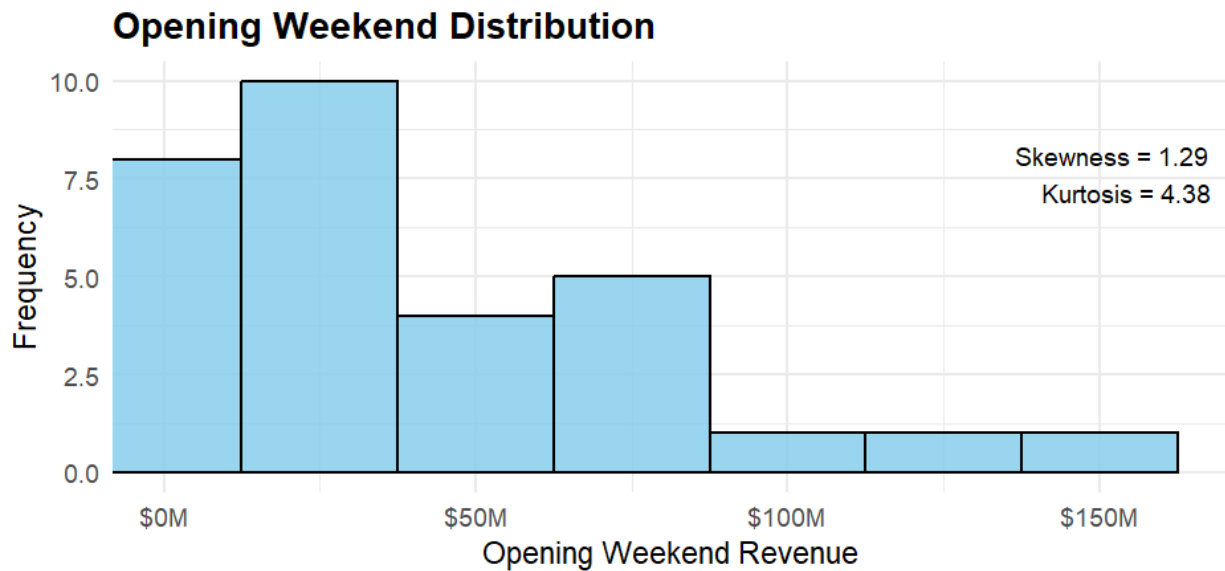


Figure 1: Opening Weekend Revenue

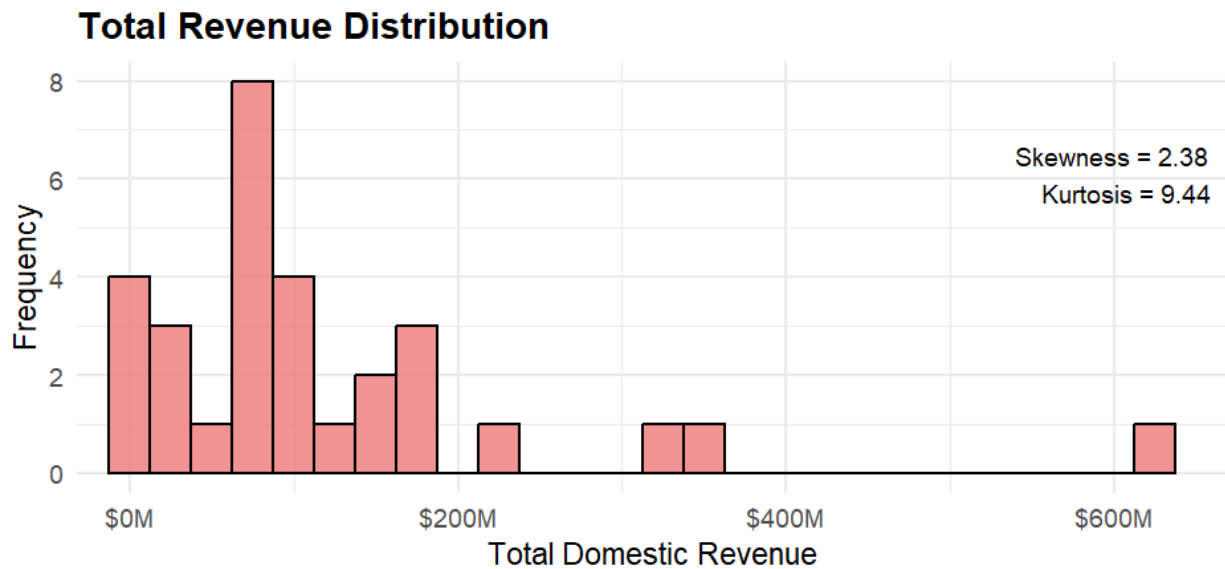


Figure 2: Total Revenue

Both opening weekend and total domestic revenue are strongly right-skewed with high kurtosis.

For opening weekend, skewness is 1.29 and kurtosis is 4.38; for total revenue, skewness is 2.38

and kurtosis is 9.44. Total revenue is noticeably more heavy-tailed than opening weekend, suggesting that a few more successful movies are extending the tail. Total revenue also accounts for a larger time period, looking at picture's whole life cycle, whereas opening weekend limits time to 3 days, limiting potential for successful films to differentiate themselves. This is consistent with De Vany and Walls (1999) and motivates the use of log transformations throughout the remainder of the analysis.

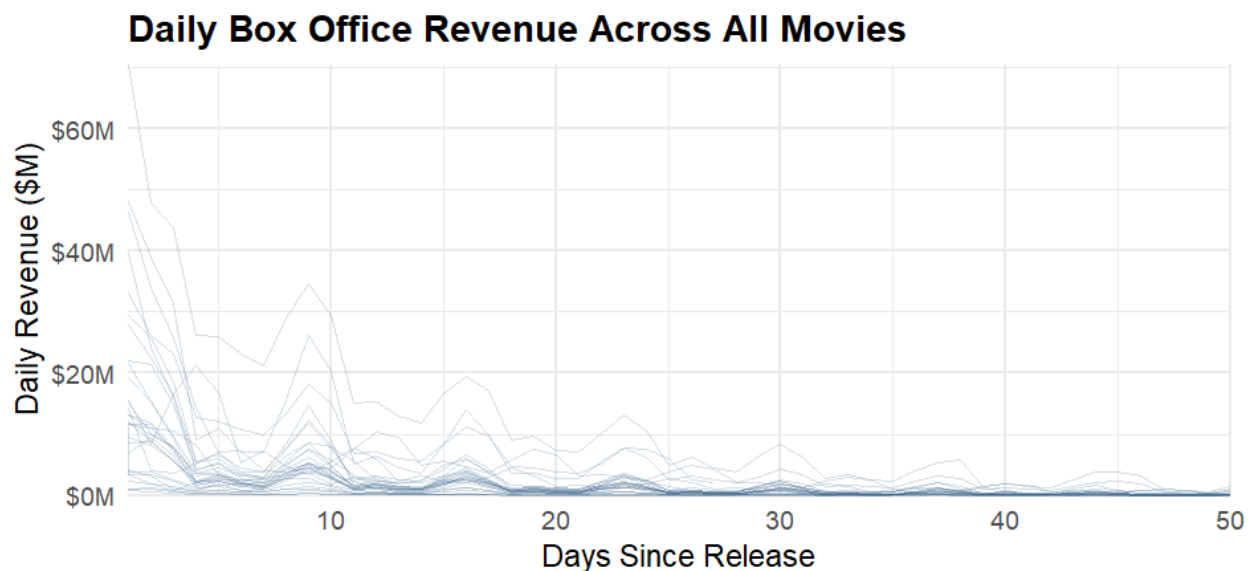


Figure 3: Daily Box Office Revenue (First 50 days)

Plotting daily revenue across all films for the first 50 days reveals two features:

- (i) a sharp downward trend following release
- (ii) a pronounced weekly pattern with weekend spikes. Both features must be addressed before time series modeling.

5. The Cross-Sectional Relationship

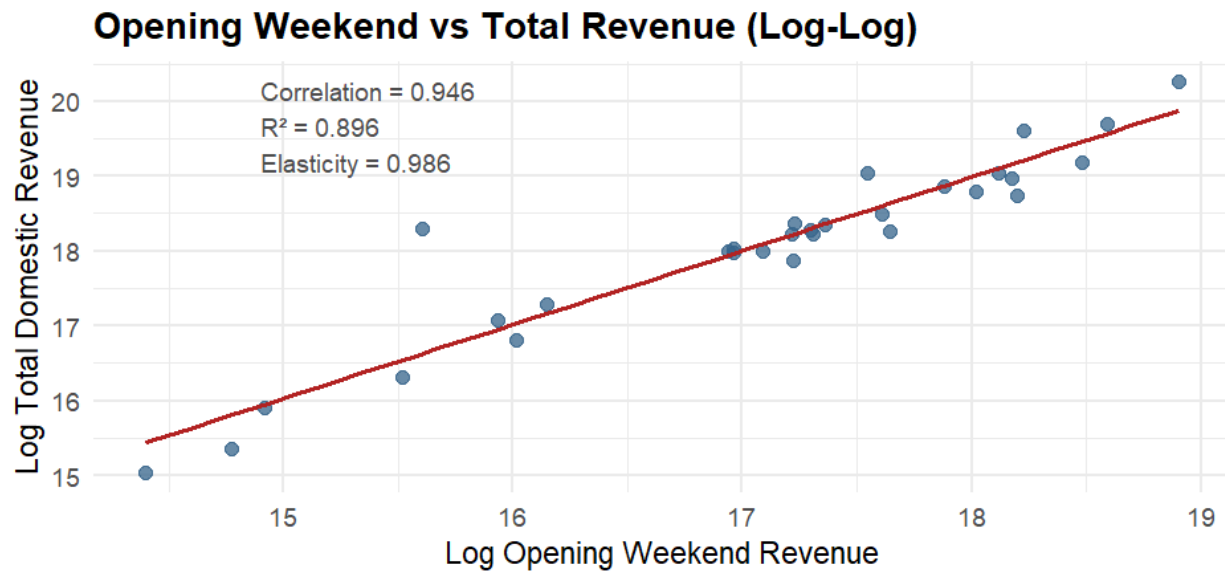


Figure 4: Opening Weekend vs Total Revenue

A log-log scatter of total revenue against opening weekend revenue is almost perfectly linear.

The correlation in logs is 0.946, the simple regression has $R^2 = 0.896$, and the estimated elasticity is 0.986. A 1% increase in opening weekend revenue is associated with approximately a 1% increase in total domestic revenue, and opening weekend alone explains roughly 90% of cross-sectional variation. This is the central piece of evidence in the paper: early performance is highly informative about final performance.

6. Cross-Sectional Model

6.1 Baseline specification

The baseline cross-sectional model is

$$\log(\text{Total}_i) = \beta_0 + \beta_1 \log(\text{Opening}_i) + \beta_2 \text{Franchise}_i + \beta_3 \log(\text{Opening}_i) \times \text{Franchise}_i + \varepsilon_i$$

The interaction term tests whether franchise films convert opening weekend into total revenue more efficiently than non-franchise films. If $\beta_3 > 0$, franchise films scale more efficiently than stand-alone films.

6.2 Results

Table 1: Regression Results

OLS Regression Results				
Dependent variable: log(revenue) · n = 30 · df = 26				
Variable	Estimate	Std. Error	t	p-value
Intercept	-1.005	1.570	-0.640	0.528
log_opening	1.133	0.097	11.715	< 0.001
franchise	2.262	2.807	0.806	0.428
log_open_fr	-0.152	0.163	-0.933	0.359
R ² = 0.911 · Adj. R ² = 0.901		Residual SE = 0.386		F(3, 26) = 88.98, p < 0.001

Opening weekend is highly significant with an elasticity of 1.13, which is slightly above one. Neither the franchise dummy nor the interaction term is statistically significant. Once opening weekend is accounted for, there is no evidence that franchise films generate higher total revenue on average, nor that they convert early revenue into total revenue more efficiently. These results provide direct answers to both research questions posed in the introduction. The main question asked to what extent early box office performance can be used to predict total domestic revenue.

The evidence here is strong: opening weekend revenue alone explains roughly 91% of the

variation in total revenue across films, the estimated elasticity is slightly above one, and the coefficient is statistically significant at the 0.1% level. Early performance is therefore an extremely informative signal about final performance. The secondary question asked whether franchise films convert opening weekend revenue into total revenue more efficiently than stand-alone films, which would appear as a positive and significant interaction term between $\log(\text{Opening})$ and the franchise dummy. The estimated interaction coefficient is small, negative (-0.15), and statistically insignificant ($p = 0.359$), so there is no support for this hypothesis in the data. Franchise status itself is also insignificant. One interpretation is that whatever advantage franchise films may have : brand recognition, an established fan base, pre-existing demand is already fully reflected in how large the film opens, leaving no residual franchise effect once opening weekend is controlled for.

6.3 Extended model with budget and runtime

Table 2: Regression Results with budget and runtime

OLS Regression Results n = 30 · df = 24				
Variable	Estimate	Std. Error	t	p-value
Intercept	-0.694	1.713	-0.405	0.689
log_opening	1.121	0.113	9.881	< 0.001
franchise	2.161	2.856	0.757	0.457
log_open_fr	-0.153	0.166	-0.921	0.366
log_budget	0.077	0.076	1.006	0.324
runtime	-0.003	0.003	-0.795	0.435
R ² = 0.915 · Adj. R ² = 0.898		Residual SE = 0.392		F(5, 24) = 51.86, p < 0.001

Budget and runtime are both insignificant, and adding them barely moves R^2 ($0.911 \rightarrow 0.915$). This is a meaningful result: variables such as budget, which one might expect to strongly influence total revenue, are already reflected in opening weekend performance. Once opening weekend is included, they add no incremental explanatory power.

7. Predictive Power of Different Early Windows (Prof. Feedback)

A natural follow-up question is whether the first week alone carries the predictive power, or whether weeks 2 and 3 are similarly informative. To address this, I estimate separate log-log regressions of total revenue on Day 1, Week 1, Week 2, and Week 3 revenue, and then combine all three weeks in a single specification.

Table 3: Regression Results with week (1,2,3) and First Day

Predictor	Elasticity (β)	R^2	p-value
Day 1 revenue	0.99	0.890	< 0.001
Week 1 revenue	1.05	0.943	< 0.001
Week 2 revenue	1.05	0.987	< 0.001
Week 3 revenue	0.78	0.947	< 0.001

All four windows are strong predictors individually. Notably, Week 2 revenue is the single strongest in-sample predictor ($R^2 = 0.987$). A plausible interpretation is that Week 1 figures are partly inflated by pre-release marketing and advance ticket sales, whereas by Week 2 word-of-mouth has taken effect and the signal is cleaner. The four estimates are nonetheless close, so in practice Week 1 (or opening weekend alone) is sufficient for a reasonable forecast.

Table 4: Regression Results with week (1,2,3) in one model

Combined Model — OLS Regression Results				
$\log(\text{total gross}) \sim \log(w1) + \log(w2) + \log(w3) \cdot n = 28 \cdot df = 24$				
Variable	Estimate	Std. Error	t	p-value
Intercept	0.826	0.454	1.818	0.082
$\log(w1)$	0.325	0.063	5.135	< 0.001
$\log(w2)$	0.567	0.125	4.533	< 0.001
$\log(w3)$	0.139	0.069	2.011	0.056
$R^2 = 0.994 \cdot \text{Adj. } R^2 = 0.993$		Residual SE = 0.104		$F(3, 24) = 1267, p < 0.001$

When all three weeks are included jointly, Week 1 and Week 2 are both highly significant ($p < 0.001$), Week 3 is marginal ($p \approx 0.056$), and the combined R^2 rises to 0.994. Week 3 therefore contributes little incremental information once the first two weeks are known, supporting the view that the bulk of predictive information is delivered in the first two weeks of a film's theatrical run.

8. Time Series Analysis of Daily Revenue

8.1 Why I don't difference the data

The daily series is clearly non-stationary, exhibiting both a strong downward trend and a weekly seasonal pattern. The standard approach would be to difference the series, but two issues arise:

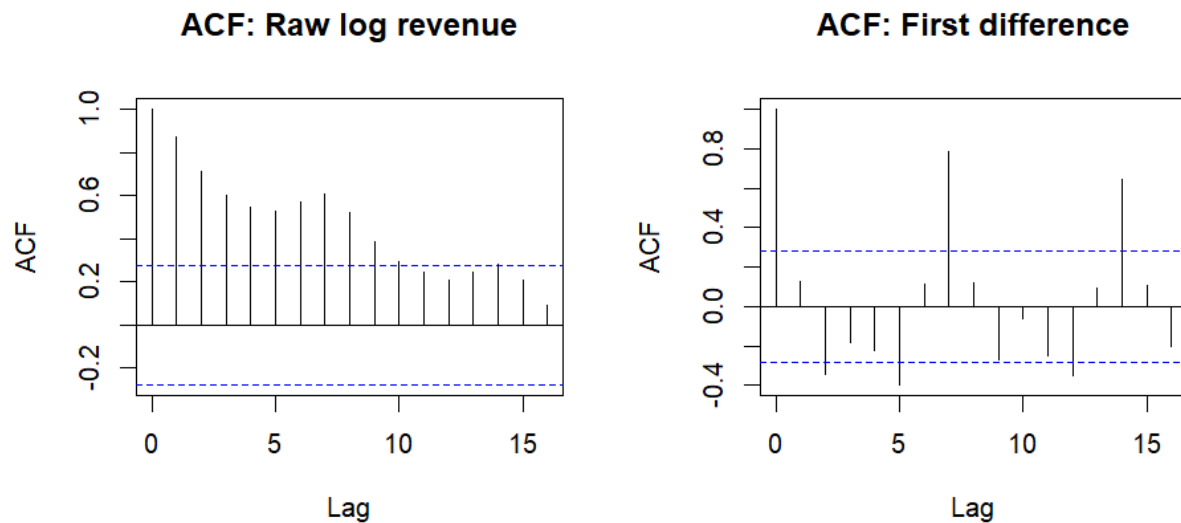


Figure 5: ACF plots of Daily Data

1. Differencing inflates variance. The variance of the log of average daily revenue is 1.42, the variance of its first difference drops to 0.17, but the variance of the second difference rises again to 0.28. Because the decay is approximately exponential, repeated differencing introduces noise into what is fundamentally a smooth deterministic pattern.

2. The trend is informative. The downward trend reflects systematic audience decay, not random drift, and differencing would discard informative structure.

Rather than differencing, I therefore model trend and seasonality directly via regression and treat the residuals as the stationary component, which is then modeled with AR/ARMA specifications.

ACF plots of the raw log series and its first difference confirm that even after first differencing there is still significant autocorrelation at the weekly lag.

8.2 Detrending regression

The time series regression is

$$\log(G_t) = \alpha + \delta t + \sum_{d=1}^6 \gamma_d D_d + \varepsilon_t$$

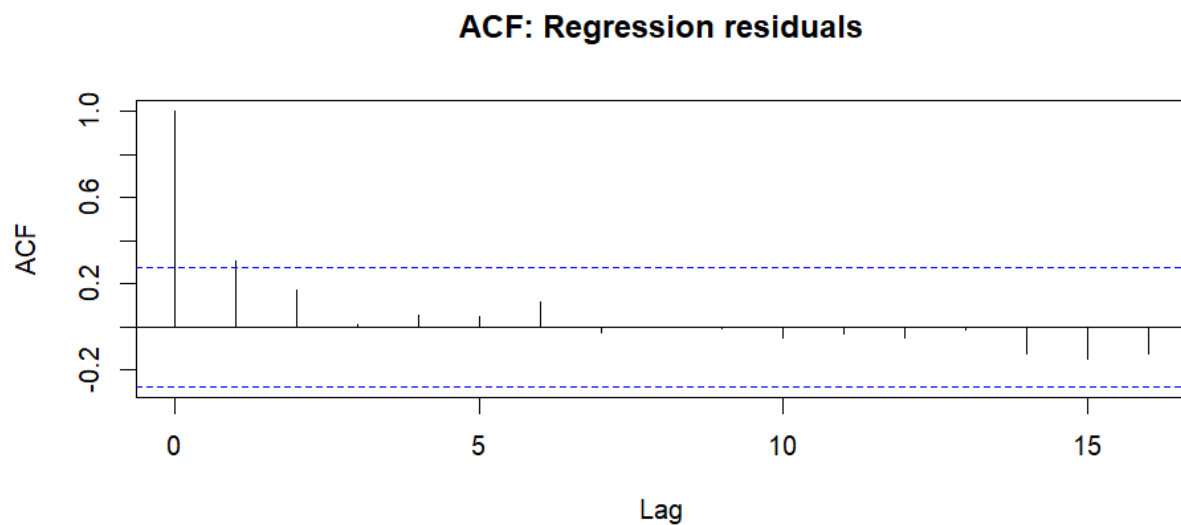
where D_d are day-of-week dummies. This is distinct from the cross-sectional regression in Section 6: it operates on the time dimension (average daily revenue across films), not across films. The estimated trend coefficient is 0.088 (about an 8.8% decay per day in log terms over the first 50 days), and the regression explains a high share of the variation in daily revenue.

Daily Decay & Day-of-Week Model — OLS Regression Results				
log(gross) ~ days + dow · n = 50 · df = 42				
Variable	Estimate	Std. Error	t	p-value
Intercept	16.144	0.063	254.498	< 0.001
days	-0.074	0.001	-51.611	< 0.001
dow1 (Mon)	0.341	0.076	4.502	< 0.001
dow2 (Tue)	0.136	0.076	1.799	0.079
dow3 (Wed)	-0.453	0.076	-5.987	< 0.001
dow4 (Thu)	-0.387	0.076	-5.112	< 0.001
dow5 (Fri)	-0.627	0.076	-8.278	< 0.001
dow6 (Sat)	-0.754	0.076	-9.957	< 0.001
R ² = 0.987 · Adj. R ² = 0.985		Residual SE = 0.146		F(7, 42) = 459.8, p < 0.001

8.3 Residual diagnostics

Residual Autocorrelation — Ljung-Box Test				
Testing for serial correlation in regression residuals				
Test	Lags	χ^2	df	p-value
Box-Ljung	14	9.215	14	0.817
No significant autocorrelation detected in residuals ($p = 0.817 > 0.05$)				

After removing trend and seasonality, the residuals are substantially closer to stationary. The residual ACF no longer exhibits weekly spikes, and the Ljung–Box test at 14 lags yields $Q = 9.22$, $p = 0.817$, so I cannot reject the null of no autocorrelation. The deterministic structure has been successfully removed.



8.4 Modeling residual dynamics

I fit several candidate models on the residuals and compared them with AIC and BIC.

Model	AIC	BIC
AR(1)	-61.00	-55.27
AR(2)	-59.22	-51.58
MA(1)	-58.61	-52.87
ARMA(1,1)	-59.16	-51.51
ARMA(2,1)	-58.35	-48.79
ARMA(1,2)	-57.66	-48.10
auto.arima	-62.92	-59.10
Lower values indicate better fit · Best per criterion highlighted		

Both AIC and BIC point to a simple AR(1) specification; auto.arima independently selects AR(1) but reports slightly better AIC/BIC values than the manually fit AR(1). This is not because it's a different model — both are ARIMA(1,0,0) — but because auto.arima detected that the residuals already have a mean of approximately zero (since they come from a regression, this is true by construction) and therefore fit the model **without an intercept term**. The manually fit `arima(y, order=c(1,0,0))` includes an intercept by default, so it estimates one extra parameter. Fewer parameters means a smaller penalty in AIC and BIC, which is why auto.arima looks slightly better on paper. Functionally the two models produce essentially identical forecasts. I report the manually fit AR(1) in the rest of the paper because it keeps the comparison with the other candidate models (AR(2), MA(1), ARMA(1,1), etc.) consistent — all of those are also fit with intercepts, so comparing them against an auto.arima version without an intercept would not be a fair like-for-like ranking. The estimated AR(1) coefficient is $\phi \approx 0.47$, indicating moderate persistence in daily revenue after trend and seasonality have been removed.

8.5 Combined model

Combining the components, the final time series specification is

$$\log(G_t) = \alpha + \delta t + \sum_{d=1}^6 \gamma_d D_d + u_t, \quad u_t = \phi u_{t-1} + \varepsilon_t$$

This integrates the structural component (trend and seasonality) with the stochastic component (AR(1) persistence) in a unified specification.

9. Out-of-Sample Forecasting

To evaluate the combined model out-of-sample, I train it on the first 40 days of two randomly selected films and forecast days 41–50. Rather than committing to AR(1) based on in-sample AIC/BIC alone, I estimate the full hybrid model for every candidate residual-dynamics specification from Section 8.4 and compare their forecast performance directly. The two films selected by the random seed were Shazam! Fury of the Gods and M3GAN.

Table 5. Forecast errors on log daily revenue by model and film:

Model	Movie	RMSE	MAE
Regression + AR(1)	Shazam! Fury of the Gods	0.446	0.368
Regression + AR(1)	M3GAN	0.289	0.215
Regression + AR(2)	Shazam! Fury of the Gods	0.512	0.384
Regression + AR(2)	M3GAN	0.280	0.194
Regression + MA(1)	Shazam! Fury of the Gods	0.471	0.414
Regression + MA(1)	M3GAN	0.289	0.217
Regression + ARMA(1,1)	Shazam! Fury of the Gods	0.495	0.378
Regression + ARMA(1,1)	M3GAN	0.284	0.194
Regression + ARMA(2,1)	Shazam! Fury of the Gods	0.511	0.392
Regression + ARMA(2,1)	M3GAN	0.290	0.218
Regression + ARMA(1,2)	Shazam! Fury of the Gods	0.569	0.431
Regression + ARMA(1,2)	M3GAN	0.284	0.202

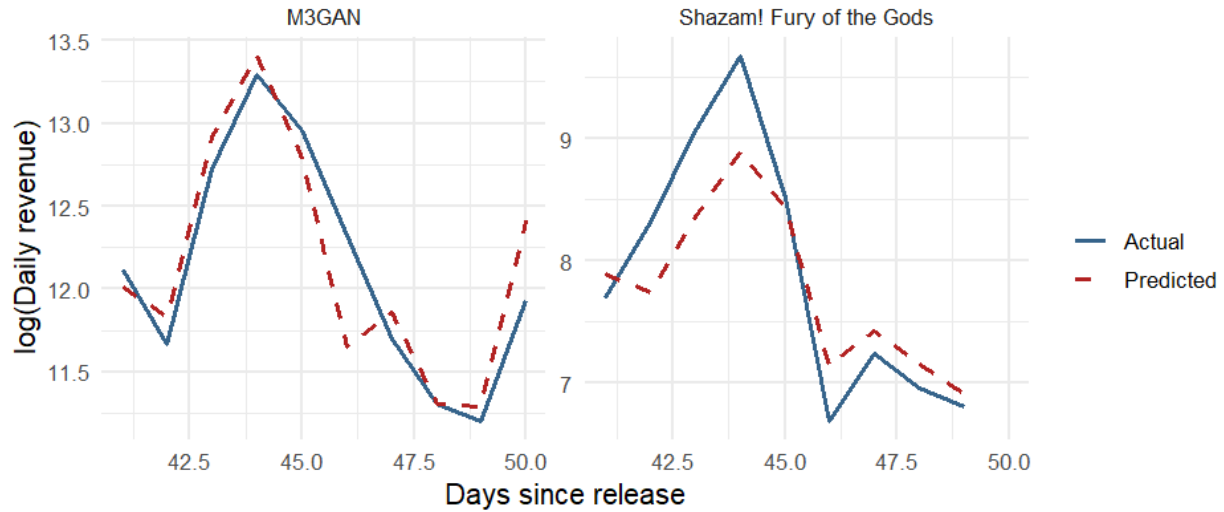
Table 6. Average forecast performance across films, ranked by RMSE:

Average forecast performance across films		
Ranked by RMSE		
Model	Avg RMSE	Avg MAE
Regression + AR(1)	0.368	0.292
Regression + MA(1)	0.380	0.315
Regression + ARMA(1,1)	0.389	0.286
Regression + AR(2)	0.396	0.289
Regression + ARMA(2,1)	0.401	0.305
Regression + ARMA(1,2)	0.426	0.316
Best model highlighted · Lower values indicate better forecast accuracy		

The hybrid Regression + AR(1) specification achieves the lowest average RMSE across the two test films, confirming the in-sample AIC/BIC ranking with independent out-of-sample evidence. ARMA(1,1) attains a marginally lower average MAE (0.286 vs 0.292), but the differences are small and AR(1) dominates on RMSE, which penalises larger errors more heavily. Importantly, more flexible specifications (AR(2), ARMA(2,1), ARMA(1,2)) perform worse out-of-sample despite estimating additional parameters, consistent with mild overfitting on the 40-day training window.

M3GAN is forecast noticeably more accurately than Shazam! Fury of the Gods across every candidate model, consistent with M3GAN's status as a smaller, more stable long-tail performer; Shazam declined unusually rapidly after its opening weekend. The forecast plot (Section 9) shows that under the winning AR(1) specification the model captures the downward trend and weekly peaks in both cases, which is the primary requirement of a simple time series model in this setting. Taken together, the in-sample information criteria and the out-of-sample forecast comparison both point to the same conclusion: a parsimonious Regression + AR(1) hybrid is sufficient, and the additional complexity of higher-order ARMA specifications is not warranted on this data.

Forecast vs Actual (log daily revenue) — Regression + AR(1)



10. Conclusion

Opening weekend revenue is an extremely strong predictor of total domestic box office revenue for 2023 films, explaining approximately 91% of cross-sectional variation with an elasticity close to unity. Franchise status, production budget, and runtime add no meaningful explanatory power once opening weekend is included, consistent with the view that these variables are already reflected in how large a film opens. Decomposing the early weeks, Week 1 and Week 2 are both highly predictive while Week 3 contributes little incremental information. Predictive power is therefore concentrated in the first half of a film's theatrical run, consistent with industry intuition.

On the time series side, the raw daily revenue series exhibits both trend and seasonality and should not be differenced: variance rises between the first and second differences. Removing trend and seasonality via regression leaves residuals that pass the Ljung–Box test, and AR(1) is preferred by both AIC and BIC for the residual dynamics. This in-sample ranking is confirmed out-of-sample: among six candidate hybrid specifications evaluated on two held-out films,

Regression + AR(1) achieves the lowest average RMSE, and higher-order ARMA models perform worse despite their additional flexibility.

11. Limitations and Future Work

I want to briefly acknowledge that I would have liked to include a larger sample of films across multiple years for this project, and I apologize that I was not able to do so within the time available. The single-year sample is the main limitation of the analysis below, and extending it is the most natural next step.

A primary limitation is sample size and scope: 30 films from a single year (2023). A larger sample spanning multiple years (e.g., 2022 and 2024) would permit stronger inference and would allow testing whether the results are stable across different box office environments; 2022, for instance, was still recovering from the pandemic. Potentially useful additional controls include marketing spend, critic and audience scores, and genre fixed effects. Finally, the forecast evaluation uses only two films; a full cross-validation exercise across all films would provide a more reliable estimate of out-of-sample performance.

12. AI Usage

AI was used to read this paper for fixing formatting and correcting grammar. It was also used to produce the visuals all the tables and math equations

References

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